Sun, Oct 4 at 12:29 PM

You said:



Don’t give answer yet but I’m confused since usually MLE gives us data. I know that the Yi distribution is 0.5 chances of being 0 and 0.5 chances of being 1. Can I assume the data is 5 of 0 and 5 of 1 and use that to make mle? Usually the likelyhood only be calculated with real data

12:29 PM

ChatGPT said:

Yes — your instinct that an MLE is usually computed from observed data is correct. The key point here is that the problem is asking you to derive the MLE **symbolically as a function of whatever data you observe**, not to plug in a made-up sample.

Also, one important correction: you **cannot say Y_i has probability 0.5 of being 0 or 1** unless $\theta = 0$.

You have

$$X_i \sim N(\theta, 1), \qquad Y_i = I\{X_i > 0\}.$$

So

$$P(Y_i = 1) = P(X_i > 0).$$

Since X_i is centered at θ, this probability depends on θ. For example:

- if $\theta = 0$, then yes, $P(X_i > 0) = 0.5$;
- if $\theta = 2$, then $P(X_i > 0)$ is close to 1;
- if $\theta = -2$, then it is close to 0.

That probability is exactly what the problem calls

$$\psi = P(Y_1 = 1).$$

So you should think of

$$Y_i \sim \text{Bernoulli}(\psi),$$

where ψ is unknown.

Now suppose the actual observed data happened to be something like

$$y_1 = 1, \quad y_2 = 0, \quad y_3 = 1, \quad y_4 = 1, \ldots$$

Then you would write the likelihood using those observations. But since the problem gives no numerical dataset, keep the observations general:

$$y_1, \ldots, y_n.$$

For one observation,

$$P(Y_i = y_i) = \psi^{y_i}(1-\psi)^{1-y_i}.$$

Therefore the likelihood for the whole sample has the form

$$L(\psi) = \prod_{i=1}^{n} \psi^{y_i}(1-\psi)^{1-y_i}.$$

I'll stop there so I don't give you the MLE itself yet.

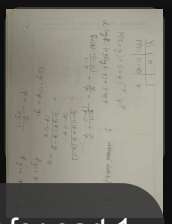
The main conceptual distinction is:

> **Data:** $y_1, \ldots, y_n$, whatever values were actually observed.
> **Parameter:** ψ, an unknown probability.
> **MLE:** a function of the observed y_i's that chooses the ψ making those observations most likely.

So no, you should **not assume 5 zeros and 5 ones** unless the problem explicitly tells you that was the observed sample. If it did say "we observed 5 zeros and 5 ones," then you could plug those counts into the likelihood.

12:29 PM

You said:



Is this the correct answer for part 1

12:48 PM

ChatGPT said:

You are very close in setup, but **this is not yet the correct answer for part (a)**. The main issue is that you wrote the likelihood for **one observation** Y_i, while the problem gives $Y_1, \ldots, Y_n$, so the MLE has to use the whole sample.

Your Bernoulli model is correct:

$$P(Y_i = y_i) = (1-\psi)^{1-y_i}\psi^{y_i}.$$

For the whole dataset,

$$L(\psi) = \prod_{i=1}^{n}(1-\psi)^{1-y_i}\psi^{y_i}.$$

Take logs:

$$\ell(\psi) = \sum_{i=1}^{n}\left[(1-y_i)\log(1-\psi) + y_i \log \psi\right].$$

Now differentiate:

$$\ell'(\psi) = -\frac{\sum_{i=1}^{n}(1-y_i)}{1-\psi} + \frac{\sum_{i=1}^{n} y_i}{\psi}.$$

Since

$$\sum_{i=1}^{n}(1-y_i) = n - \sum_{i=1}^{n} y_i,$$

you get

$$-\frac{n-\sum y_i}{1-\psi} + \frac{\sum y_i}{\psi} = 0.$$

Now I would recommend that **you solve this last equation yourself** before I give the final MLE.

One useful way to see what went wrong in your page: because you differentiated the likelihood of just **one Bernoulli observation**, your calculation is essentially asking, "If I observed $y_i = 1$, what value of ψ makes that single observation most likely?" Of course it pushes toward $\psi = 1$; if $y_i = 0$, it pushes toward $\psi = 0$.

But with n observations, the zeros and ones balance each other, and that's where the meaningful MLE comes from.

12:48 PM

You said:

For the second question I want to get distribution of psi but I am stuck at P(psi=a) is probability of probability of xi greater than 0. It's a nested probability can you give me some hints

01:22 PM

ChatGPT said:

The key conceptual point is: ψ **itself is not random here.** It is an unknown parameter, so you do **not** want to calculate something like

$$P(\psi = a).$$

What is random is your **estimator** $\hat{\psi}$, because it depends on the random sample $X_1, \ldots, X_n$.

For part (b), think in this order.

From part (a), you should get an estimator $\hat{\psi}$ that depends on $\hat{\theta}$. First find the relationship between ψ and θ:

$$\psi = P(X_i > 0).$$

Standardize $X_i \sim N(\theta, 1)$:

$$P(X_i > 0) = P\left(\frac{X_i - \theta}{1} > \frac{0 - \theta}{1}\right).$$

So try simplifying that probability using the standard normal CDF Φ. You should end up with a function

$$\psi = g(\theta).$$

Then your MLE has the form

$$\hat{\psi} = g(\hat{\theta}).$$

Now for the confidence interval, don't ask

$$P(\psi = a).$$

Instead ask:

> **What is the approximate sampling distribution of $\hat{\psi}$?**

You already know something very useful:

$$\bar{X} \sim N\left(\theta, \frac{1}{n}\right).$$

And your $\hat{\psi}$ is a function of $\bar{X}$:

$$\hat{\psi} = g(\bar{X}).$$

So this is exactly where the **delta method** comes in. If

$$\sqrt{n}(\bar{X} - \theta) \xrightarrow{d} N(0, 1),$$

then

$$\sqrt{n}\left(g(\bar{X}) - g(\theta)\right) \xrightarrow{d} N\left(0, [g'(\theta)]^2\right).$$

Since $g(\theta) = \psi$, this gives you the approximate distribution of

$$\hat{\psi} - \psi.$$

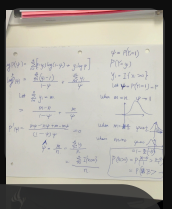
So your next two steps are just:

1. Find $g(\theta)$ from $P(X > 0)$.
2. Compute $g'(\theta)$.

Don't build a distribution for ψ. Build the **sampling distribution of $\hat{\psi}$**. That's the distinction that unlocks part (b).

01:22 PM

You said:



Is this correct for part a

01:46 PM

ChatGPT said:

Yes — your **part (a) derivation is correct now**.

You correctly wrote

$$\ell(\psi) = \sum_{i=1}^{n} \left[(1 - y_i) \log(1 - \psi) + y_i \log \psi \right],$$

then differentiated:

$$\ell'(\psi) = \frac{\sum_{i=1}^{n} (y_i - 1)}{1 - \psi} + \frac{\sum_{i=1}^{n} y_i}{\psi}.$$

Letting

$$m = \sum_{i=1}^{n} y_i,$$

you got

$$\ell'(\psi) = \frac{m - n}{1 - \psi} + \frac{m}{\psi}.$$

Setting that equal to zero gives

$$\hat{\psi} = \frac{m}{n} = \frac{1}{n} \sum_{i=1}^{n} Y_i.$$

So the final MLE is

$$\boxed{\hat{\psi} = \bar{Y}}.$$

That is exactly right.

One small notation comment: on your page you wrote something like $p'(\psi)$. Since you differentiated the **log-likelihood**, it is cleaner to write

$$\ell'(\psi)$$

instead.

Also, your intuition sketches on the right are good: if all $Y_i = 1$, then $\hat{\psi} = 1$; if half are 1, then $\hat{\psi} = 1/2$; if all are 0, then $\hat{\psi} = 0$.

So part (a) is done correctly.

01:46 PM

Mon, Oct 5 at 3:37 PM

You said:

Don’t give me hint about2 yet tell me what delta method is and give me a short example

03:37 PM

ChatGPT said:

The **delta method** is a way to get the approximate distribution of a function of an estimator.

Suppose

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \sigma^2).$$

Now suppose you care not about θ itself, but about

$$g(\theta).$$

Then for a smooth function g,

$$\sqrt{n}\big(g(\hat{\theta}) - g(\theta)\big) \xrightarrow{d} N\left(0,\, [g'(\theta)]^2 \sigma^2\right).$$

The intuition comes from a first-order Taylor expansion:

$$g(\hat{\theta}) \approx g(\theta) + g'(\theta)(\hat{\theta} - \theta).$$

So the variability of $g(\hat{\theta})$ is approximately the variability of $\hat{\theta}$, multiplied by $g'(\theta)$.

A short example: suppose

$$X_1, \ldots, X_n \sim N(\mu, 1),$$

so

$$\bar{X} \sim N\left(\mu, \frac{1}{n}\right).$$

Let

$$g(\mu) = \mu^2.$$

Then

$$g'(\mu) = 2\mu.$$

By the delta method,

$$\sqrt{n}(\bar{X}^2 - \mu^2) \xrightarrow{d} N(0, 4\mu^2).$$

Equivalently, for large n,

$$\bar{X}^2 \approx N\left(\mu^2, \frac{4\mu^2}{n}\right).$$

So the basic recipe is:

$$\text{distribution of } \hat{\theta} \quad \longrightarrow \quad \text{differentiate } g \quad \longrightarrow \quad \text{approximate distribution of}$$

It is essentially **Taylor expansion + asymptotic normality**.

03:37 PM

You said:

I have not learned the expression Theya minus theta hat -> normal under d can you teach me

04:17 PM

ChatGPT said:

Yes. The notation is just a compact way of saying **the distribution of an estimator becomes approximately normal as the sample size gets large**.

Suppose $\hat{\theta}$ estimates some fixed unknown parameter θ. Then

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \sigma^2)$$

has three pieces.

First,

$$\hat{\theta} - \theta$$

is the **estimation error**. For example, if the true value is $\theta = 5$ and your estimate is $\hat{\theta} = 5.2$, then the error is 0.2.

Second, the symbol

$$\xrightarrow{d}$$

means **converges in distribution**. Roughly:

> As $n \to \infty$, the distribution of the expression on the left looks more and more like the distribution on the right.

So

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \sigma^2)$$

means that for large n,

$$\sqrt{n}(\hat{\theta} - \theta) \approx N(0, \sigma^2).$$

The third piece is the $\sqrt{n}$. Why is it there? Usually, as n increases, $\hat{\theta}$ gets closer to θ. Its error often shrinks at rate

$$\frac{1}{\sqrt{n}}.$$

So if we didn't multiply by $\sqrt{n}$,

$$\hat{\theta} - \theta \to 0.$$

Its distribution would just collapse toward zero. Multiplying by $\sqrt{n}$ "magnifies" that shrinking error so that we can see its limiting shape.

A very familiar example is the sample mean. Suppose

$$X_1, \ldots, X_n$$

are iid with

$$E[X_i] = \mu, \qquad \mathrm{Var}(X_i) = \sigma^2.$$

We estimate μ using

$$\hat{\mu} = \bar{X}.$$

We know

$$E[\bar{X}] = \mu, \qquad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

Therefore

$$\bar{X} - \mu$$

has standard deviation

$$\frac{\sigma}{\sqrt{n}}.$$

Multiply by $\sqrt{n}$:

$$\sqrt{n}(\bar{X} - \mu).$$

Its variance becomes

$$n\frac{\sigma^2}{n} = \sigma^2.$$

And the Central Limit Theorem tells us

$$\boxed{\sqrt{n}(\bar{X} - \mu) \xrightarrow{d} N(0, \sigma^2)}$$

or equivalently,

$$\boxed{\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \xrightarrow{d} N(0, 1).}$$

You may actually recognize this better in the more familiar form

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right).$$

These are basically saying the same thing. Starting from

$$\sqrt{n}(\bar{X} - \mu) \approx N(0, \sigma^2),$$

divide by $\sqrt{n}$:

$$\bar{X} - \mu \approx N\left(0, \frac{\sigma^2}{n}\right),$$

then add μ:

$$\boxed{\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right).}$$

So if the $\xrightarrow{d}$ notation is unfamiliar, for now you can mentally translate

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \sigma^2)$$

as

$$\boxed{\hat{\theta} \text{ is approximately } N\left(\theta, \frac{\sigma^2}{n}\right) \text{ for large } n.}$$

That is the form we'll need to understand the delta method.

04:17 PM

Sat, Oct 10 at 11:26 AM

You said:

Could u give me the hint of question part b again

11:26 AM

ChatGPT said:

Yes. For part **(b)**, the goal is to build an approximate 95% CI for ψ, and the clean route is:

$$X_i \sim N(\theta, 1), \qquad \psi = P(X_i > 0).$$

Your first task is to express ψ as a function of θ. Standardize:

$$\psi = P(X_i > 0) = P\left(\frac{X_i - \theta}{1} > -\theta\right).$$

So rewrite that in terms of the standard normal CDF Φ. Call the resulting function

$$\psi = g(\theta).$$

Next, because for the normal model

$$\hat{\theta} = \bar{X},$$

the MLE invariance principle gives

$$\hat{\psi} = g(\hat{\theta}) = g(\bar{X}).$$

Now use the fact that

$$\bar{X} \sim N\left(\theta, \frac{1}{n}\right).$$

Since $\hat{\psi} = g(\bar{X})$, apply the delta method:

$$\mathrm{Var}(\hat{\psi}) \approx [g'(\theta)]^2 \, \mathrm{Var}(\bar{X}).$$

So your next concrete step is:

$$\boxed{\text{Find } g(\theta) \text{ and then calculate } g'(\theta).}$$

Once you have that, you'll have

$$\hat{\psi} \approx N\left(\psi, \frac{[g'(\theta)]^2}{n}\right).$$

Then the usual 95% CI template is

$$\hat{\psi} \pm 1.96 \times \mathrm{SE}(\hat{\psi}).$$

I'd stop there for now. Try finding $g(\theta)$ and $g'(\theta)$, and send me what you get.

11:26 AM